The Journal of Emergency Medicine, Vol. 64, No. 5, pp. 564–573, 2023
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0736-4679/$ - see front matter

https://doi.org/10.1016/j.jemermed.2023.02.011

Original Contributions

The Impact of Peripheral Intravenous Catheter Insertion Technique and Catheter-to-Vein Ratio on Postinsertion Failure

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☐ **Abstract—Background: Peripheral intravenous catheter (PIVC) postinsertion failure rates are unacceptable. Ultrasonography is an adjunctive tool that may improve PIVC utilization success. Objectives: Determine if ultrasonographically guided (USG) PIVCs placed in the emergency department (ED) significantly decreases postinsertion failure rate, increases utility time, and decreases postremoval complication rate. Determine if catheter-to-vein ratio (CVR) predicts postinsertion failure. Methods: Participants were randomized to either standard or USG cohort. Data collection included participant and PIVC characteristics, vein measurements, postinsertion failure, and postremoval complication. Chi-square analysis compared postinsertion failure rates. Group *t*-test compared utility times. Postremoval complication rates were compared with standard rate analysis. The receiver operating characteristic curve was calculated to determine if CVR could predict postinsertion failure. An enrollment of 582 was estimated. Results: A total of 223 patients were enrolled, with 222 PIVCs investigated. Standard cohort included 116 PIVCs and USG cohort included 106 PIVCs. A total of 212 vein diameters were analyzed. USG PIVC insertion did not result in fewer postinsertion failures ($p = 0.654$) or longer utility time ($p = 0.808$). Postremoval complications were not significantly lower ($p = 0.414$). Receiver operating characteristic curve showed a cut-off CVR of ~ 0.21. Area under the curve was 0.621 ($p = 0.063$, 95% confidence interval 0.508–0.734). Conclusion: The USG technique did not decrease postinsertion failure rate, increase utility time, or significantly decrease postremoval complication rate. A CVR predictive of postinsertion failure could not be determined.** **© 2023 Elsevier Inc. All rights reserved.**

☐ **Keywords—peripheral; intravenous; catheter; postinsertion; failure; ultrasonographically guided; catheter-to-vein ratio**

Introduction

Peripheral intravenous catheter (PIVC) insertion is a common emergency department (ED) intervention used for blood draws, diagnostic laboratory studies, and for administering intravenous fluids, medications, blood components, and nutrition. An established PIVC is subject to failure as a result of various complications such as pain, phlebitis, infiltration, leakage, occlusion of catheter or vein, and infection, particularly cellulitis (1–5). These complications may lead to more serious sequelae such as impaired circulation, tissue necrosis, and systemic infection (3). Any of these complications alone or in combination results in premature removal of the PIVC in most cases (1–5). Postinsertion failure of a PIVC has direct adverse effects on the patient, which include, but are not limited to, the need to replace the PIVC, further patient discomfort, venous depletion, and longer hospital stay. Any of these adverse effects may decrease patient satisfaction and increase cost (3). Diagnostic studies and treatment, in the ED or continued during in-patient care,

RECEIVED: 26 September 2022; FINAL SUBMISSION RECEIVED: 23 December 2022;
ACCEPTED: 17 February 2023

often rely on the integrity of a PIVC inserted by ED staff. Considering the insertion risks of PIVCs, evaluation of survivorship is important. Postinsertion failure rates of PIVCs have been a concern for decades. Despite the gain of evidence that has led to change in practice, the percentage of overall PIVC failures continues to range between 18.8% and 77% (1–4).

Postinsertion failure rates of PIVCs inserted in the ED using the ultrasonographically guided (USG) technique are not well known. There is a paucity of data in the published literature comparing postinsertion failure rates and longevity between PIVCs inserted in the ED using the standard (landmark) technique and PIVCs inserted using the USG technique. Only one study performed a similar comparison, and it was limited to having only ED technicians place PIVCs and 72 h of dwell time studied (6).

No studies to date have investigated ED PIVC postremoval complications and whether or not catheter-to-vein ratio can predict postinsertion failure. This study is the next step toward understanding the outcomes of PIVCs placed in the ED with regard to insertion technique, catheter-to-vein ratio, and postremoval complications. This study had four objectives: 1) determine if USG PIVC insertion results in lower postinsertion failures compared with PIVCs inserted using the standard technique; 2) determine if USG PIVC insertion results in longer utility time compared with the standard technique; 3) determine if USG PIVC insertion results in lower postremoval (up to 96 h) complications compared with the standard technique; and 4) determine if catheter-to-vein ratio can predict all-cause postinsertion failure (excluding accidental removal).

Materials and Methods

Study Design and Setting

Our study was a prospective, randomized, controlled trial conducted at a single tertiary hospital. The ED has an annual census of 134,000 patients, with an additional 2,500 trauma activations. Twenty percent of the population is pediatric. Nurses and paramedics were the primary source for PIVC insertion, with additional support from ED staff (nurses, paramedics, and physicians) trained in USG PIVC insertion.

Trained investigators screened potential participants, obtained consent, performed measurements, and collected data. Data were collected and managed using the secure electronic data collection web application Research Electronic Data Capture (7,8). The institutional review board approved the study protocol (IRB 042-19), and the study was registered in ClinicalTrials.gov (NCT04218643).

Investigator Training

Prior to data collection, the ability to obtain accurate and consistent vein diameter measurements was confirmed. The 12 investigators who performed vein diameter measurement completed a 1-h training session comprised of a didactic portion and a skills laboratory. The investigators were trained using both SonoSite Edge II and X-Porte, each equipped with the same model 13-6 MHz, 6-cm linear probe (FUJIFILM SonoSite, Inc. Bothell, WA). Internal consistency, inter-rater reliability, and test-retest reliability analysis was completed for the 12 investigators. For internal measurement consistency of each investigator, the calculated Cronbach's alpha score of > 0.97 was considered excellent. For inter-rater reliability among the 12 investigators, the calculated intraclass correlation coefficient of > 0.9 was considered excellent. For test-retest reliability of each investigator, the calculated Pearson correlation coefficient for the vertical measurement was > 0.9, and for the horizontal measurement was > 0.84, both considered acceptable. Please refer to Appendix A (available online) for a more detailed description of vein diameter measurement training.

Enrollment

A convenience sample was acquired from the study population of adult patients registered in the ED. Trained investigators communicated with ED nurses and providers to initially identify and screen potential participants to confirm that they met all criteria for enrollment. Inclusion criteria were: age ≥ 18 years; no existing PIVC; an order for a PIVC or treatment requiring a PIVC; and acceptable vital signs not requiring emergent resuscitation. Exclusion criteria were: prisoners; patients who were institutionalized; patients triaged as myocardial infarction alert, stroke alert, or trauma alert; and patients unable to provide consent. Patients were not enrolled if their primary nurse or provider declined based on clinical instability. We excluded pediatric patients due to lack of resources and narrow variability in vein size and utility. Patients with an existing PIVC were also not included in our study for the purpose of avoiding vein depletion, a factor that can contribute to postinsertion failure (3,4). Enrollment was completed after reviewing inclusion and exclusion criteria, explaining the study, answering all questions, and obtaining written consent.

Intervention, Measurement, and Data Collection

Randomization occurred after consent was obtained. The control cohort consisted of participants with a PIVC inserted using the standard technique, and the intervention cohort consisted of participants with a PIVC inserted

using the USG technique. Randomization was computer generated and then the results transferred to blank standard data collection forms that were individually sealed in an envelope; each envelope was numbered in sequence and retrieved in sequence. After randomization, the investigator coordinated with the patient's nurse to have the PIVC inserted by a qualified ED staff member. In the event that a PIVC could not be successfully inserted according to randomization (e.g., difficult access, delay of ultrasound resources), the participant received vascular access as per the treatment team. To limit bias, ED staff members were not provided with the specifics of our study and were instructed to place the PIVC the same as for any patient.

After the PIVC was inserted, the investigator performed vein diameter measurements with one of the ultrasound machines used for vein measurement training. The vein diameter used for data analysis was the vertical axis plus the horizontal axis divided by two. The vertical axis was measured from the most superior point of the lumen border to the most inferior point of the lumen border, and the horizontal axis was measured from the most left lateral point of the lumen border to the most right lateral point of the lumen border. The vertical and horizontal axis measurements were obtained just proximal to the catheter tip. The vein diameter was excluded from analysis if the investigator could not obtain vein diameter measurements before the PIVC was utilized for an infusion or the investigator was unable to measure both vertical and horizontal axes (e.g., poor vein structure or ultrasound image quality). Data collected at time of PIVC insertion included participant characteristics (Table 1), PIVC insertion and site characteristics (Table 1), and vein diameter measurements.

The electronic health record (EHR) of participants that did not have their study PIVC removed in the ED was monitored until the study PIVC was identified as removed. The participant's EHR was then reviewed to obtain data pertaining to utility, survivability, and removal reason. In our study, postinsertion failure was identified when any of the following reasons for PIVC removal was entered into the participant's EHR: leakage, drainage, phlebitis, painful, occluded, infiltrated, and catheter damage. Accidental dislodgement was not included as failure, because the securing method and maintenance of PIVCs were not a focus of our study. When more than one reason for removal was entered, the EHR was further investigated, with a second investigator to determine the primary reason for removal. If the EHR did not provide sufficient data, an interview with the participant or the participant's nurse was attempted to complete data collection. If the PIVC did not fail, a trained investigator attempted to contact the participant between 48 and 96 h after the PIVC was removed. Upon contact, the investigator inquired if any postremoval complications developed. If the participant was not contacted, the participant's EHR was reviewed to identify any subsequent ED or primary care physician visits related to postremoval PIVC complications.

Statistical Analysis

The baseline failure rate for postinsertion PIVCs at our study site was 30%. We had set an a priori improvement of an absolute percentage change of 10% as significant. Using a single tail change in improvement ($\alpha = 0.025$) with the intervention with an 80% power, the number needed for each cohort was calculated to be 291 subjects.

Random number assignments were performed using 'R' (R Core Team, 2013. R: A language and environment for statistical computing) to allocate the standard and USG cohorts. The 'runs test' available in SPSS v28.0 (IBM, Armonk, NY) was used to validate randomness of the random number series. A conservative p-value of 0.50 was used to determine if the sequence of numbers was random. Randomization continued until a sequence with a p-value > 0.50 was obtained. Once completed, the random number sequence was saved to an Excel file (Microsoft Corporation, Redmond, WA) and utilized for randomization.

Descriptive analysis was performed for PIVC characteristics and variables within each cohort. Inferential statistics were performed for the comparisons between the two cohorts. For postinsertion failure rates, chi-square analysis was used. Utility times were analyzed by group t-test. Postremoval complication rates were analyzed by standard rate analysis. Postremoval complication rates were calculated as the number of complication divided by the number of patients. An a priori p-value less than 0.050 was used to determine statistical significance between cohorts.

Catheter-to-vein ratio was calculated by dividing the vein diameter by the outside diameter of the intravenous catheter. To determine a catheter-to-vein ratio predictive of failure, measurements were aggregated from the two cohorts, and a receiver operating characteristic (ROC) curve was calculated using the catheter-to-vein ratio as the independent variable and the failure (0, 1) as the dependent variable. Logistic regression analysis was also performed. Due to the exploratory nature of these analyses there was no correction for multiple comparisons.

Results

Screening, enrollment, and PIVCs analyzed are shown in Figure 1. A total of 745 registered ED patients were screened for participation, of which 223 were enrolled.

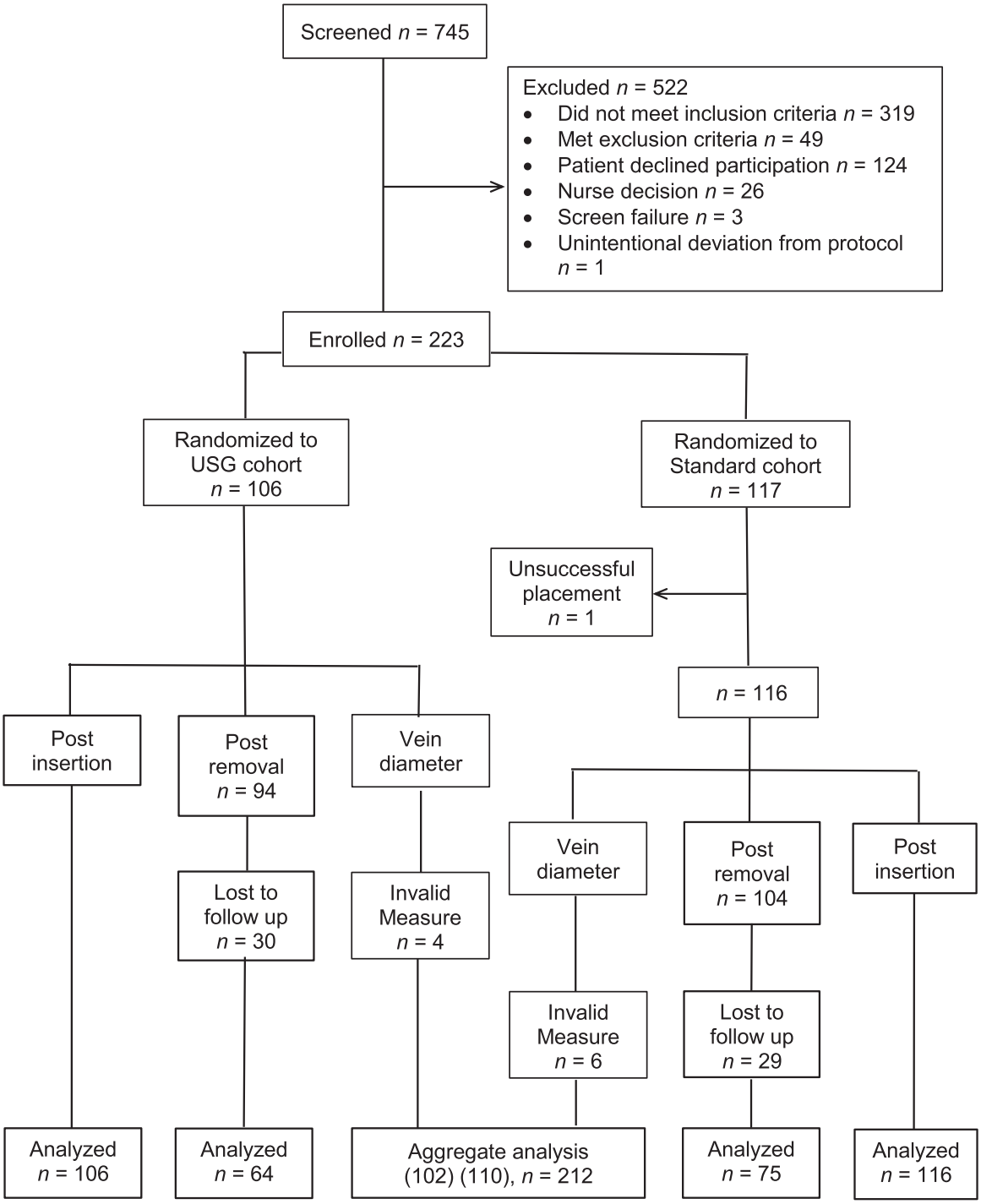

Figure 1. Flowchart of screening, enrollment, and peripheral intravenous catheters analyzed. USG = ultrasonographically guided

A total of 106 PIVCs were randomized to the USG technique cohort (intervention) and analyzed. A total of 117 PIVCs were randomized to the standard technique cohort (control), with 116 analyzed due to one PIVC insertion being unsuccessful. Distribution difference between the cohorts was considered good at < 10%. After adjusting for PIVCs lost to follow-up, PIVCs removed without complication and used for postremoval analysis were 64 in the USG cohort and 75 in the standard cohort. After subtracting invalid vein diameters (no measurement or measurement after infusion), an aggregate total of 212 vein diameters were used for further analysis.

Demographic data are presented in Table 1. Patients in the standard cohort were older, with a lesser proportion

Table 1. Participant and Peripheral Intravenous Catheter Placement Characteristics

Characteristic	Overall (n = 222)	USG (n = 106)	Standard (n = 116)
Age, years: mean (range)	53 (19–95)	50 (19-92)	55 (21–95)
Female, n (%)	121 (54.5%)	60 (56.6%)	61 (52.6%)
Race, n (%)			
White	198 (89.2%)	98 (92.5%)	100 (86.2%)
Black	11 (5%)	4 (3.8%)	7 (6%)
Other	13 (5.9%)	4 (3.8%)	9 (7.8%)
No medical condition, n (%)	106 (47.7%)	57 (53.8%)	49 (42.2%)
Catheter size, n (%)			
16 gauge	1 (0.5%)	1 (0.9%)	0 (0%)
18 gauge	40 (18%)	23 (21.7%)	17 (14.7%)
20 gauge	162 (73%)	80 (75.5%)	82 (70.7%)
22 gauge	18 (8.1%)	2 (1.9%)	16 (13.8%)
24 gauge	1 (0.5%)	0 (0%)	1 (0.9%)
Anatomical position, n (%)			
Hand	11 (5%)	1 (0.9%)	10 (8.6%)
Forearm	111 (50%)	70 (66%)	41 (35.3%)
Antecubital fossa	100 (45%)	35 (33%)	65 (56%)
Vein diameter cm, range (mean)	0.07–0.73 (0.28)	0.11–0.73 (0.28)	0.07–0.62 (0.28)

USG = ultrasonographically guided.

of females. Most participants in our convenience sample identified themselves as white. Participants with no medical condition and no history of i.v. drug use were slightly fewer overall than participants with a medical condition and history of i.v. drug use.

Data describing PIVC characteristics are also presented in Table 1. Among all PIVCs, the 20-gauge catheter was most commonly used for placement (73%), with 4.8% more placed in the USG cohort than the standard cohort. The 18-gauge catheter was the second most placed (18%), with 7% more placed in the USG cohort than the standard cohort. The 22-gauge catheter was the third most placed (8.1%), with 11.9% more in the standard cohort than in the USG cohort. The 24- and 16-gauge catheters were placed the least. Overall, the anatomical locations chosen for PIVC placements were nearly equally distributed between the forearm (50%) and antecubital fossa (45%), with the remaining PIVCs placed in the hand. The forearm was chosen 30.7% more in the USG cohort compared with the standard cohort. The antecubital fossa was chosen 23% more in the standard cohort compared with the USG cohort. The hand was chosen only in the standard cohort. The mean vein diameter among all PIVCs and in both cohorts was the same (0.28 cm). PIVCs were placed in a range of larger veins (0.11–0.73 cm) in the USG cohort compared with the standard cohort (0.07–0.62 cm). Additional data describing postinsertion failure and postremoval complication characteristics can be reviewed in Appendix B (available online).

Comparing the USG technique with the standard technique, the USG technique did not result in fewer postinsertion failures (11.3% USG vs. 9.5% standard; $p = 0.654$). Utility time (hours) was not longer when using the USG technique (mean = 23.5, SD = 33.8 USG vs. mean = 24.7, SD = 38.8 standard; $p = 0.808$). Postremoval complications were not significantly lower when using the USG technique (9.4% USG vs 12% standard; $p = 0.414$). Comparison results are shown in Table 2.

The ROC curve showed a cut-off ratio of ~ 0.21. The area under the curve was considered poor, with a result of 0.621 ($p = 0.063$, 95% confidence interval [CI] 0.508–0.734). Logistic regression attempting to predict postinsertion failure was not significant ($p = 0.079$) and did not identify any additional patients. In addition, model fit was poor (H-L statistic 0.108, with an R^2 of only 3.5%; Nagelkerke-0.035). The ROC curve is displayed in Appendix C (available online).

Discussion

Our study provides methodology and data that will be beneficial for future development of more robust studies and improving PIVC postinsertion outcomes. We found

Table 2. Comparison Results

Comparison	USG	Standard	*p* Value
Postinsertion failure, n (%)	12 (11.3%)	11 (9.5%)	0.654
Utility time hours, mean (SD)	23.53 (33.8)	24.7 (38.8)	0.808
Postremoval complication, n (%)	6 (9.4%)	9 (12%)	0.414

USG = ultrasonographically guided.

that utilizing ultrasonography to place PIVCs did not decrease postinsertion failures, increase utility times, or significantly decrease postremoval complications. With the exception of accidental dislodgement, our analysis to identify a catheter-to-vein ratio that predicts all-cause postinsertion failure was inconclusive, requiring further investigation.

The postinsertion failure rate for participants admitted to an in-patient room in our study was 27.8%, which is comparable with other studies. Postinsertion failure rates among admitted patients reported by Marsh et al. and Shokoohi et al. were 32% and 25%, respectively (2,6). The study performed by Shokoohi et al. showed no significant difference in postinsertion failure rates between the standard and USG techniques, with reported rates of 21% (95% CI 15–28%) and 27% (95% CI 21–35%), respectively (6). Our data supported this insignificant difference with a failure rate of 11.3% using the USG technique, and a failure rate of 9.5% using the standard technique. These data may justify considering the use of the USG technique sooner or even immediately in some circumstances.

Miliani et al. reported a postremoval PIVC-related adverse event rate of 21.7% and did not report if any participants returned for treatment of a postremoval adverse event (5). Our study had no participants return for treatment of a postremoval complication, however, the overall postremoval complication rate was only 10.8%. The lower percent could be a result of our study, including only PIVCs placed in the ED.

The mean vein diameter in the USG cohort was the same as the standard cohort of 0.28 cm. This is unusual because ED staff trained to use the USG technique can typically find larger veins and choose them for better insertion success. One study that investigated USG PIVCs placed in the ED setting reported that 49% of PIVCs were placed in veins with a diameter of 0.4 cm or more, and 63% were placed proximal to the antecubital fossa (9). Two factors in our study might have contributed to USG PIVCs placed in veins with diameters smaller than historically reported. Standard of practice at our site was to place PIVCs below the upper arm, thus substantially decreasing access to larger veins. Our study site also has a high volume of ED admissions, thereby creating more highly experienced USG-trained staff who have greater confidence in placing PIVCs in smaller veins to save time.

An unintentional finding worth noting is the considerable difference in the anatomical location chosen for PIVC placement between the two cohorts. The standard of practice at our study site was to consider placing the PIVC in the forearm prior to placing in any other anatomical location. The USG cohort had 30.7% more PIVCs placed in the forearm than the standard cohort and 23% fewer PIVCs placed in the antecubital fossa. These data suggest that the USG technique may be considered for better adherence to standard of practice and not just for difficult peripheral intravenous access. This is a focus for further investigation.

Limitations

A primary limitation of our study is that the calculated sample size for a power of 80% could not be attained. The unforeseen COVID-19 pandemic had a detrimental affect on our study site, and as a result, data collection became crucially limited. Contributing factors were a significant decline in each of the following: ED census, access to patients, clinical staff, investigators, and possibly, patient interest.

The practice of routinely placing PIVCs below the upper arm for our single-site study limited access to larger veins. As a result, the range of data for statistical analysis regarding catheter-to-vein ratio was limited. The practice also limits the generalizability of the data to patients with PIVCs placed below the upper arm. No PIVC was placed above the elbow for our study. The protocol for our study enabled the investigators to enroll predominately lower-acuity patients with lesser haste for PIVC placement. Data from our study are not representative of the entire population of adult ED patients.

Predominantly, ED staff members were trained for vein diameter measurement. Certified sonographers were not used because our investigation team determined that using certified sonographers for routine vein measurement is not feasible in the ED. This decision was supported with the results from the study performed by Sharp et al., which demonstrated that nurses who place peripherally inserted central catheters are able to measure vein diameter with precision compared with a sonographer (10). Our study

demonstrated that ED staff members with various levels of training provide precise and consistent vein measurements when using ultrasonography.

Conclusions

Our data showed that PIVCs placed in the ED using the USG technique did not decrease postinsertion failure rate, increase utility time, or significantly decrease postremoval complication rate compared with the standard (landmark) technique. Excluding accidental removal, a catheter-to-vein ratio predictive of postinsertion all-cause failure could not be determined. Further investigation with a larger sample size and fewer limitations is needed to gain stronger, statistically significant evidence.

Appendix

Appendix A: Vein Measurement Training and Data Collection to Determine Precision

Investigators were instructed to measure vein diameter using the following steps: 1) Confirm that if an automatic blood pressure cuff is on the patient's arm, that it is not inflated; 2) Adjust the depth to the shallowest setting; 3) Apply a generous amount of sterile gel to the area proximal to the i.v. site; 4) Using transverse view, locate the tip of the catheter, then move the transducer slightly proximal until the tip of the catheter disappears so that it doesn't interfere with measurement; 5) Lift the transducer until the view is unclear and then very lightly press down until the vein returns into view clearly, but without compressing the vein; 6) Freeze the view; 7) If the view could be better, repeat 5 and 6; 8) Using the calipers, measure the vertical axis and horizontal axis of the lumen of the vein.

A male volunteer consented to provide 12 measurement points on his arms, which included at least two points in the hand, forearm, and antecubital fossa. An experienced technician made sure each measurement point provided a consistent and clear view of the vein lumen. Measurement points were marked using a surgical skin marker and given a unique identifier for recording. Each investigator measured the vertical axis and horizontal axis of the vein, which were then recorded. After no less than a 30-min break and without access to the first set of measurements, each investigator returned to repeat the measurements. Every measurement was performed in the Emergency Department and with the male volunteer in the same body position.

Appendix B: Descriptive Data of Postinsertion Failures and Postremoval Complications

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Failure Characteristics	Overall (n = 23)	USG (n = 12)	Standard (n = 11)
Catheter size, n (%)			
18 gauge	6 (26.1%)	2 (16.7%)	4 (36.4%)
20 gauge	14 (60.9%)	9 (75%)	5 (45%)
22 gauge	3 (13%)	1 (8.3%)	2 (18.2%)
Anatomical position, n (%)			
Forearm	11 (47.8%)	8 (66.7%)	3 (27.3%)
Antecubital fossa	12 (52.2%)	4 (33.3%)	8 (72.7%)
Vein diameter cm, range (mean)	0.14–0.5 (0.24)	0.14–0.5 (0.23)	0.15–0.4 (0.25)
Type of infusion, n (%)			
Contrast dye	8 (34.8%)	4 (33.3%)	4 (36.4%)
NSS	21 (91.3%)	12 (100%)	9 (81.8%)
D5W	7 (30.4%)	3 (25%)	4 (36.4%)
LR	2 (8.7%)	1 (8.3%)	1 (9.1%)
Potassium	2 (8.7%)	2 (16.7%)	0 (0%)
Magnesium	5 (21.7%)	3 (25%)	2 (18%)
Antibiotic	16 (69.6%)	8 (66.7%)	8 (72.7%)
Other medication	17 (73.9%)	9 (75%)	8 (72.7%)
Blood	1 (4.3%)	1 (8.3%)	0 (0%)

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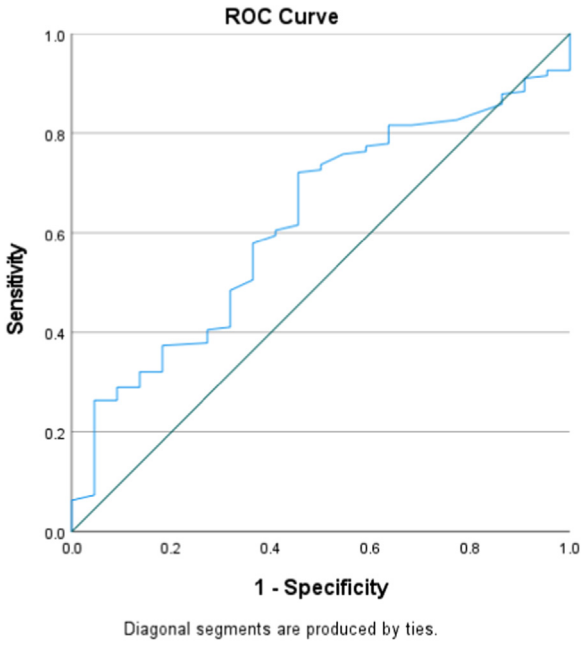
Failure Characteristics	Overall (n = 23)	USG (n = 12)	Standard (n = 11)
One type of infusion	1 (4.3%)	0 (0%)	1 (9.1%)
> 1 type of infusion	22 (95.7%)	11 (91.7%)	11 (100%)
> 2 types of infusion	15 (65.2%)	7 (58.3%)	8 (72.7%)
Unit PIVC removed, n (%)			
ED	5 (21.7%)	2 (16.7%)	3 (27.3%)
Observation	1 (4.3%)	1 (8.3%)	0 (0%)
In patient	17 (73.9%)	9 (75%)	8 (72.7%)
Removal reason, n (%)			
Leakage	7 (30.4%)	3 (25%)	4 (36.4%)
Painful	4 (17.4%)	2 (16.7%)	2 (18.2%)
Occluded	3 (13%)	1 (8.3%)	2 (18.2%)
Infiltration	9 (39.1%)	6 (50%)	3 (27.3%)
Complication Characteristics	Overall (n = 15)	USG (n = 6)	Standard (n = 9)
Redness/warmth	2 (13.3%)	2 (33.3%)	0 (0%)
Tenderness/pain	8 (53.3%)	3 (50%)	5 (55.6%)
Edema/swelling	9 (60%)	3 (50%)	6 (66.7%)
Hematoma	1 (6.7%)	0 (0%	1 (11.1%)
Fever	1 (6.7%)	0 (0%)	1 (11.1%)
> 1 complication	5 (33.3%)	2 (33.3%)	3 (33.3%)
> 2 complications	1 (6.7%)	0 (0%)	1 (11.1%)

USG = ultrasonographically guided; NSS = normal saline solution; D5W = 5% dextrose in water; LR = lactated Ringer's; PIVC = peripheral intravenous catheter; ED = emergency department.

Appendix C: Receiver Operator Characteristic (ROC) Curve for the Catheter-to-Vein Ratio



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ARTICLE SUMMARY

1. Why is this topic important?

Placement of a peripheral intravenous catheter (PIVC) is one of the most common invasive procedures performed in the emergency department (ED). Postinsertion failure rates among these PIVCs are unacceptably high. Just a small decrease in failure rates will result in a significantly large number of PIVCs that do not need to be replaced.

2. What does this review attempt to show?

This study attempts to show that PIVCs placed in the ED using the ultrasonographically guided (USG) technique significantly decrease postinsertion failure rate, increase utility time, and decrease postremoval complication rate compared with the standard (landmark) technique. This study also attempts to show that catheter-to-vein ratio can predict postinsertion all-cause failure (excluding accidental removal).

3. What are the key findings?

There was no significant difference between the two techniques with respect to postinsertion failure rate, longevity, and postremoval complication rate. Further investigation with a larger sample size and access to a wider range of vein sizes is needed to determine if catheter-to-vein ratio can predict all-cause postinsertion failure (excluding accidental removal). The USG technique may have an advantage toward adhering to standard of practice, thus being a focus for further investigation.

4. How is patient care impacted?

Preventing postinsertion failure decreases the risk of initial complications resulting in any or all of the following: patient discomfort, the need to replace the PIVC, venous depletion, longer hospital stay, and increased cost. Initial complications may lead to more rare but serious sequelae such as impaired circulation, tissue necrosis, and systemic infection.